

Instructions: This is a closed-book exam. You are not permitted to refer to any material or discuss the problem with anyone. Malpractice will be severely punished. Please mention your ROLL Number and name clearly in the answer sheet.

Duration: 75 minutes

Question 2.1 (Total 23pts). For the following distribution, find the Shannon, Huffman and Shannon-Fano-Elias codes. In each case, give the code tree (3pts each), the set of codeword lengths (2pts each), and the expected length (2pts each). Also compute the entropy of the distribution (2pts).

$$\mathcal{X} = \{a, b, c, d, e, f\}$$

and

$$p_X = [0.3, 0.2, 0.2, 0.15, 0.1, 0.05]$$

Question 2.2 (Total 10pts). What can you say about the Kraft sum of Huffman codes (2pts)? Justify (8pts)

Question 2.3 (5pts). We have seen that for any prefix-free code, $\mathbb{E}[\ell(X)] \geq H(X)$.

Suppose that the prefix-free code has Kraft sum $1 - \epsilon$, for some $0 < \epsilon < 1$. Derive a tighter lower bound on the expected length as a function of ϵ .

Question 2.4 (Total 10pts). There are six patients and it is known that exactly one of them is infected with a certain disease. Your goal is to design a sequence of (blood) tests to identify which one of them is infected. Since the number of test kits is limited, you would like to use these judiciously.

Based on patient history, you know that the infected person is patient i with probability $p_X(i)$, where

$$p_X = [0.3, 0.15, 0.1, 0.3, 0.1, 0.05].$$

1. Suppose that you test people sequentially in some order, and stop when you find the infected patient. What is the optimal order (in the sense of minimizing expected number of tests) of testing the patients to detect which person is infected (1pt)? What is the expected number of tests required in this case (2pts)?
2. Suppose that you are permitted to pool blood samples of multiple patients and feed it to the kit. For example, if you combine blood samples of patients 1, 3, 4 and test the pool, the kit will return a positive if one of 1, 3, 4 is infected, and return negative otherwise. You can sequentially test such pools and stop as soon as you find the infected patient. Design an algorithm to find the infected patient using the minimum expected number of tests (5pts). What is the expected number of tests (2pts)?